

Aby Johnny

Healthcare Software Engineer · ABDM / FHIR R4 / FastAPI · Kerala, India

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PROFESSIONAL SUMMARY

AI & Data Science graduate building production-grade healthcare infrastructure in India. Currently architecting ClaimsLens — an ABDM-compliant Health Information Provider (HIP) platform implementing FHIR R4 across the full insurance claim lifecycle, from consent management to NHCX submission. Experienced with the full ABDM HIP/HIU flow, AES-256-GCM encryption, ECDH-based secure data exchange, and cloud deployment on Azure. I work at the intersection of complex compliance requirements and real engineering constraints.

EXPERIENCE

Software Developer Intern · RecodeAI Solutions

2025 — Present

- Architecting ClaimsLens — a production ABDM-certified HIP platform for insurance claim digitization via NHCX
- Implemented FHIR R4 bundle generation for all HI types: Discharge Summary, OPConsult, Diagnostic Report, Prescription, and Immunization
- Built full ABDM consent lifecycle — grant, revoke, and expiry flows using Celery background jobs and PostgreSQL
- Integrated AES-256-GCM envelope encryption for health document storage on Azure Blob Storage with hospital/patient-isolated paths
- Implemented ECDH-based encrypted data push to Health Information Users (HIU) per ABDM spec
- Deployed on Azure using GitHub Actions CI/CD with OIDC-based auth and Managed Identity for Key Vault secrets
- Stack: FastAPI, PostgreSQL, SQLAlchemy, Celery, Pydantic v2, React, TypeScript, Docker, Azure

Java Full Stack Development · JSpiders, Kochi

Jul 2024 — May 2025

- Completed intensive full-stack training covering Core Java, OOP, Advanced Java, and Spring Boot fundamentals
- Built frontend interfaces with HTML, CSS, and JavaScript; developed and optimized SQL queries for relational databases

Data Science Intern · Teachnook

2024

- Applied ML algorithms, data visualization techniques, and data science workflows on real-world datasets

PROJECTS

ClaimsLens — ABDM HIP Platform · Production · Full-Stack

2025 — Present

- ABDM-certified HIP platform for insurance claim digitization — FHIR R4, consent management, NHCX integration
- Backend: FastAPI + PostgreSQL + SQLAlchemy + Celery | Frontend: React + TypeScript | Cloud: Azure
- ABDM M1 certification in progress; M2/M3 target milestones scoped and underway

Multi-Agent Task Automation · Crew AI · Python · LLMs

2024

- Orchestrated multiple AI agents using Crew AI for automated task execution with sequential and parallel pipelines
- Integrated external APIs for real-time data retrieval; engineered agent roles, prompts, and communication protocols

Hand Sign Recognition System · Computer Vision · Deep Learning

2023

- Real-time A–Z alphabet sign recognition using CNNs trained on diverse datasets with live camera tracking

AI Battery Fault Prediction · Predictive Analytics · IoT

2023

- Predictive maintenance system for vehicle batteries using ML algorithms achieving high-accuracy fault detection

TECHNICAL SKILLS

Backend: Python, FastAPI, SQLAlchemy, PostgreSQL, Celery, Pydantic v2, REST APIs

Healthcare / ABDM: FHIR R4, ABDM HIP/HIU flows, NRCES IG, NHCX, Consent Management, SNOMED CT, ICD-10, LOINC

Cloud & DevOps: Azure (Blob Storage, Key Vault, Managed Identity), GitHub Actions, Docker, OIDC

Frontend: React, TypeScript, Vite, HTML, CSS, JavaScript

Data & ML: Python (scikit-learn, TensorFlow, Keras, OpenCV), SQL, Data Analysis, Feature Engineering

Currently Learning: ABDM M2/M3 certification flows, SNOMED CSNatk, LOINC toolkit

EDUCATION

B.Tech in AI & Data Science · Viswajyothi College of Engineering, Vazhakulam *Aug 2020 — May 2024*

- Specialized in Artificial Intelligence & Data Science; served as Class Representative (S5, S6)
- Capstone projects in computer vision, predictive analytics, and multi-agent AI systems

Higher Secondary (Science) · St. Stephens H.S.S, Keerampara *Jul 2018 — Jun 2020*

CERTIFICATIONS

◆ Generative Models for Developers	Infosys Springboard · 2024
◆ Prompt Engineering	Infosys Springboard · 2024
◆ Introduction to OpenAI GPT Models	Infosys Springboard · 2024
◆ Data Visualization using Python	Infosys Springboard · 2024
◆ Tableau Desktop Specialist	Infosys Springboard · 2024
◆ Tableau in 7 Steps	Infosys Springboard · 2024

ADDITIONAL

Languages: English (Fluent), Malayalam (Native)

Leadership: Regional Youth President — KCYM; Class Representative, AD Department (S5, S6)

Portfolio: abyjohnny.github.io/Aby